

Generative Market Simulation — Supplement

Design Justification of the Information Environment & Related-Work Positioning

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Private, heterogeneous, decaying signals

Each round, every agent privately sees a noisy reading of each market's fundamental value p^* :

$$s = \text{clip}(p^* + \mathcal{N}(0, \sigma_t)), \quad \sigma_t = \sigma \left(1 - \rho \frac{t-1}{T-1}\right)$$

- The contract is **binary**, so this value is the event probability — a YES share pays \$1 iff YES $\Rightarrow$ value = $P(\text{YES}) = p^*$. Giving a probability signal is not an invention; it falls out of the binary payoff. [Wolfers & Zitzewitz 2004, JEP; 2006, NBER WP]
- And each agent gets its **own private signal** — the market's task is to aggregate information scattered privately across agents, so I hand out private cues and control exactly who knows what: the standard lab aggregation design. [Plott & Sunder 1982, JPE; 1988, Econometrica]
- Each signal is **truth + Gaussian noise**: σ sets how well-informed an agent is (agents differ; none sees p^*), and normal noise $\Rightarrow$ analytic Bayesian updating. [Grossman & Stiglitz 1980, AER; Hellwig 1980, JET]

Line A — evaluating LLMs on real prediction markets

PolyBench and peers put LLMs on live Polymarket data to predict / trade. [Cheng et al. 2026, arXiv; Halawi et al. 2024, NeurIPS] → prices are real but exogenous & frozen — the model only reacts, never sets the price.

Line B — agent-generated market simulation

LLM agents endogenously generate the market themselves — e.g. TwinMarket, with a social-media layer for agent interaction. [Yang et al. 2025, NeurIPS; Lopez-Lira 2025, arXiv] → but on stock / continuous-asset markets, not prediction markets — and the setup is noisy, hard to control.

Where GMS sits — the empty cell

Endogenous & agent-driven price, on a prediction market (binary now, extensible to multi-outcome), kept clean & controllable. A is real but frozen; B is endogenous but a stock market and not clean — neither occupies this cell.

The simulation in action — the round loop (Step → Round)

and an agent-by-agent walkthrough of one turn.

→ switch to the live system

1. Remove the market maker — does arbitrage emerge?

Drop the scripted automated market maker and let only LLM agents trade. Test whether they **discover and exploit arbitrage** on their own — instead of leaning on a market maker to supply liquidity and pull prices back in line.

2. Open up group features — does misbehavior emerge?

Enable the V1 tools — `transfer`, `create_account`, and a **discussion board** (agent-to-agent messaging). Watch whether agents spontaneously **post false information** (talk a market up or down) or form alliances.

Thanks!

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